

End-semester Examinations Winter Semester 2024-25
Course: NHSA 103 Understanding Human Behaviour

26th April 2025

Instructions

1. All the questions are compulsory
2. Marks are indicated against each question. Total Marks is 50
3. Write your section (A, B, C, D) in the front page of your answer sheet
3. There are four parts (A, B, C, D). All parts need to be answered separately. Start each part from a fresh page
4. If a question mentions word limit, it must be followed
5. Time limit is 2 hours

Total Marks = 12.5

PART A

1. Lily, a 34-year-old graphic designer, has not left her house in the past 9 months. She avoids malls, buses, or even the park near her home. When she contemplates going outside, her heart races, her palms sweat, and she feels like she's going to faint. She insists on having groceries delivered and refuses to even open the door unless it's absolutely necessary. In therapy, she revealed that these symptoms started after she had a panic attack at a shopping centre.

- a) Based on the given symptoms, would you classify Lily as having a mental disorder?
Justify your answer using at least three core criteria. **(3 marks)**
- b) What disorder does this presentation most closely align with? **(1 mark)**
- c) In order to diagnose any fear as phobia, what is the criteria? **(1 mark)**

2. Ryan, a 29-year-old man, is brought to a psychiatric evaluation by his family. He insists that government satellites are monitoring his brain activity and transmitting his thoughts to a secret agency. He also claims he can hear two distinct voices: one praises him for uncovering the truth, while the other threatens to punish him for speaking out. He sometimes reports smelling burning rubber when no one else notices anything unusual and refuses to eat because he believes his food is poisoned, despite no evidence.

- a) Based on this case, identify what kind of delusion and hallucination Ryan is experiencing? **(1.5 marks)**
- b) Based on the above case, highlight the key differences between delusions and hallucinations. **(2 marks)**

3. Which disorder is marked by instability in mood, identity and impulse control? What are the specific symptoms of this disorder? (1+3 marks)

Total Marks = 12.5

PART B

1. Define group and explain any 2 theories of group antagonism. Outline atleast 2 strategies to reduce group antagonism (4+3 marks)
2. Define attitude. With the help of any 2 theories explain the process of attitude formation. (5.5 marks)

Total Marks = 12.5

PART C

1. a) Differentiate between neuron cells and glial cells. (1 mark)
b) Name the two divisions of the autonomic nervous system and explain. (2 marks)
2. a) Amit refuses to swim in the ocean because he recently saw a viral video of a shark attack, even though such incidents are extremely rare. Identify the heuristic and explain. (0.5 + 1 marks)
b) Explain functional fixedness (1 mark)
3. What do you mean by "hemispheric specialisation"? (1 mark)
4. Limbic system regions help guide control of the internal environment, and the hypothalamus is one such region. Name and mention the main function of the three output pathways of the Hypothalamus to affect changes in the internal state. (3 marks)
5. Write a short note on the parts (at least 3 parts) of the brainstem and their functions. (3 marks)

Total Marks = 12.5

Part - D

1. Given the following situations, identify the type of memory involved: (0.5x3 = 1.5 marks)
- a) A person rides a bicycle effortlessly, but cannot explain how they do it.
- b) After several months, a person still remembers the capital cities they memorized for a quiz.
- c) A student recognizes a multiple-choice answer they had studied earlier, even though they couldn't recall it before.

2. Explain the differences between sensory memory and short-term memory in terms of duration, capacity, and function. Use examples to support your answer. (Word Limit: 100 Words max) **(3 Marks)**

3. Answer the following questions: (maximum word limit = 25 words) **(1x4 = 4 marks)**

- Some subway riders participating in a study were told to talk to strangers on their commute, while other participants were instructed not to. Then, at the end of the commute, the researchers conducting the study asked people how happy they were. Which is the independent variable and which is the dependent variable?
- Researchers visited a children's play area, where they observed how much caregivers used their digital devices. The researchers recorded any instances of caregivers yelling at the children as they played. Which is the independent variable and which is the dependent variable?
- A health psychologist measures participants' body mass index (BMI) by dividing their weight in kilograms by their height in meters squared (kg/m^2). What is the scale of measurement for BMI, and why?
- A survey collects data on participants' birth months to analyse seasonal trends in depression. What is the scale of measurement for the birth months, and why?

4. Answer the following questions: (maximum word limit = 50 words) **(2x2 = 4 marks)**

- A psychologist is studying the effect of social support on academic stress among college students. They define social support as "the perception of being cared for and having assistance available from others," and measure it using the Multidimensional Scale of Perceived Social Support (MSPSS). Academic stress is defined as "the pressure students feel due to academic workload and performance expectations," and is measured using a self-reported Academic Stress Inventory. Identify the construct and operational definitions for both variables.
- In a nutrition study, participants are grouped based on their daily sugar intake level (Low, Medium, High), and their blood glucose level (mg/dL) is measured using a glucose monitor. Identify the independent variable and its scale of measurement, and the dependent variable and its scale of measurement.